GearLoan

An equipment lending app with a full DevOps pipeline: CI/CD, security gates, Terraform on AWS, Kubernetes, GitOps and monitoring

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Lending club gear without a paper register

1. **Items**

   cameras, laptops, audio, lights, tripods, with condition and serial number

2. **Loans**

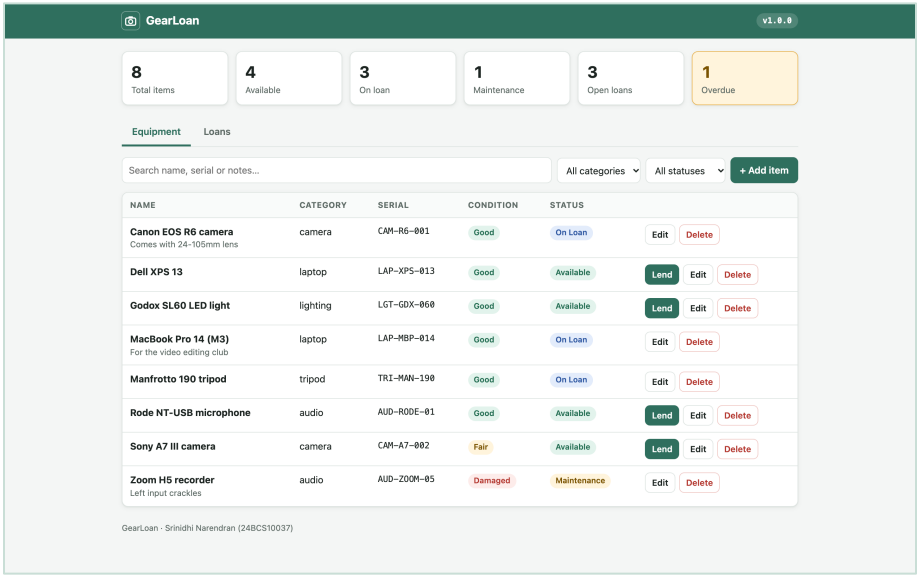
   lend with a due date, return, overdue shown automatically

3. **Rules**

   no double lending · max 3 open loans per person · damaged → maintenance

4. **API**

   FastAPI REST with GET, POST, PUT and DELETE, plus health, ready and metrics endpoints

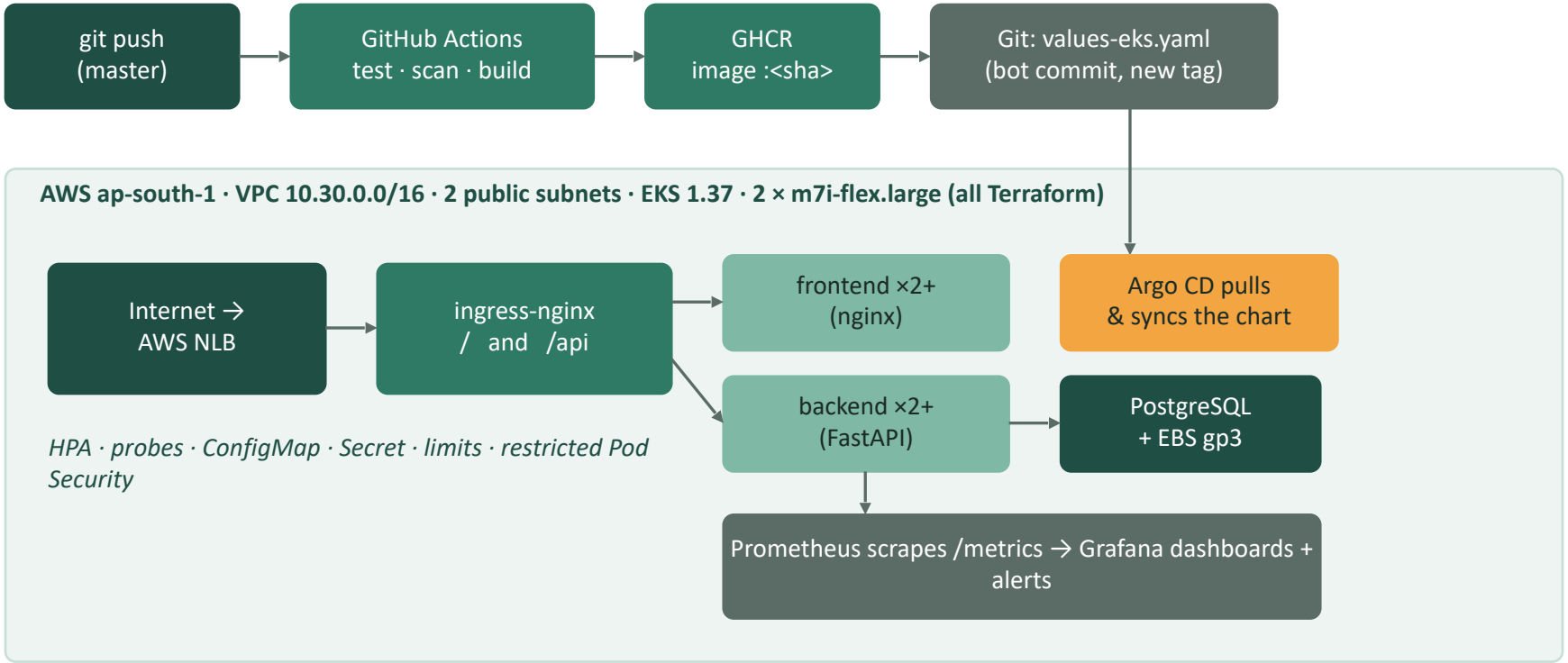


PostgreSQL 17 + Alembic migration · React UI, works on phone and desktop

Tech stack

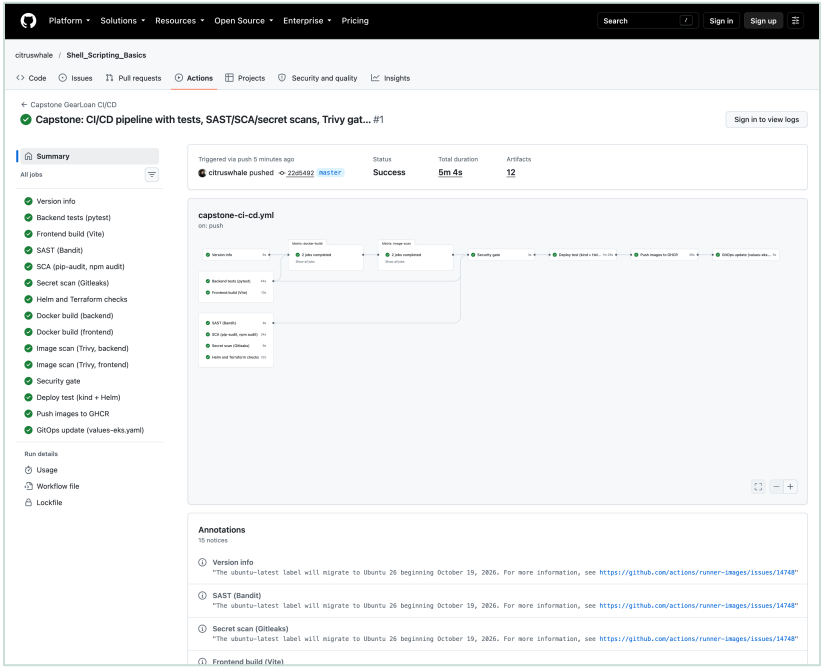
App	Python 3.12 · FastAPI · SQLAlchemy · Alembic · React 18 + Vite
Data	PostgreSQL 17 (StatefulSet + EBS volume)
Containers	Docker multi-stage, non-root · Docker Compose
CI/CD	GitHub Actions · GHCR · kind deploy test
Security	Bandit · pip-audit · npm audit · Gitleaks · Trivy
Cloud	Terraform · AWS VPC · EKS 1.37 · EBS CSI
Kubernetes	Helm 4 chart · ingress-nginx · HPA · Argo CD
Observability	Prometheus · Alertmanager · Grafana

Architecture



CI/CD pipeline on every push

1	**Test + build**	pytest on a Postgres service · frontend build
2	**Scan code**	Bandit · pip-audit · npm audit · Gitleaks · helm/terraform checks
3	**Build once**	both images, linux/amd64, saved as tarballs
4	**Trivy gate**	fail on HIGH / CRITICAL
5	**Deploy test**	kind cluster + Helm + smoke test
6	**Ship**	push the scanned images to GHCR :<sha> · bot updates Git



Run 38964516947: 15 jobs green in 5m 4s

Security gates before anything ships

0

HIGH or CRITICAL CVEs with a fix, in both images

Trivy scanned Debian 13.7 (backend) and Alpine 3.24 (frontend) plus every Python and Node package. Clean today does not mean safe forever, so it runs on every push.

SAST	Bandit on the backend code: no issues
SCA	pip-audit + npm audit: 0 known vulnerabilities
Secrets	Gitleaks over the whole Git history: no leaks
Containers	non-root users (uid 10001 / 101), read-only root filesystem
Least privilege	workflow permissions per job; no passwords in Git

Infrastructure as code with Terraform

45

resources in the plan

2

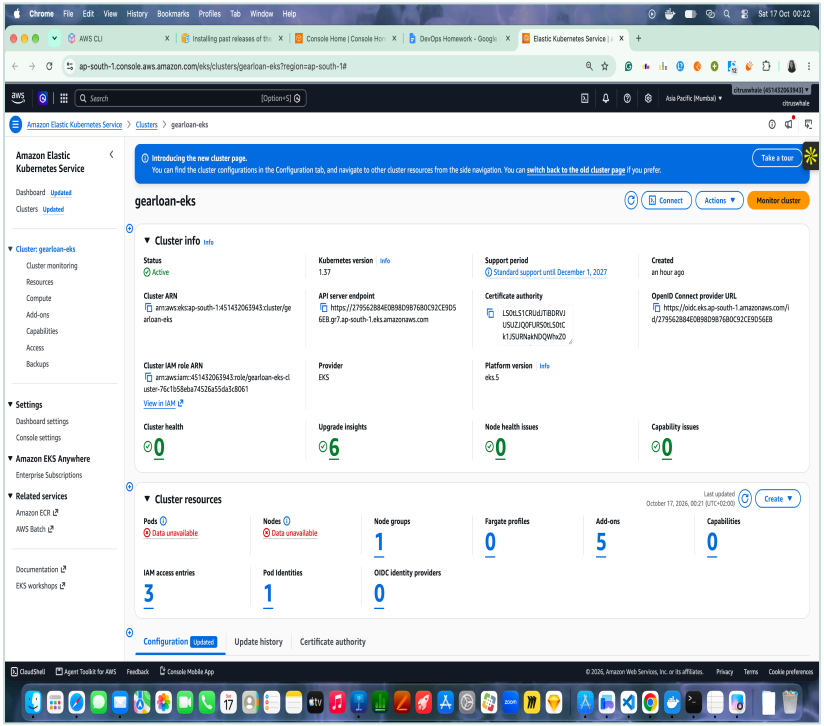
public subnets, 2 AZs

2

m7i-flex.large workers

$0.33

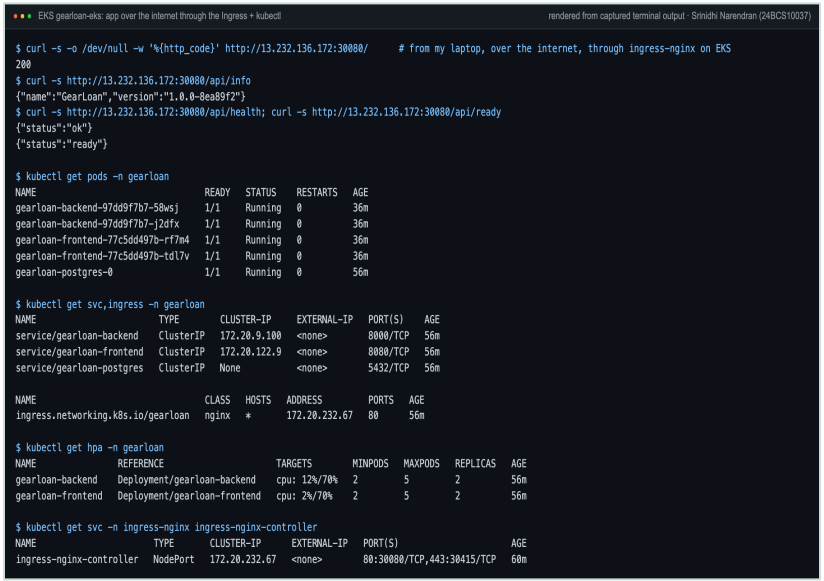
per hour, then destroy



- VPC 10.30.0.0/16 with an internet gateway, no NAT (keeps cost low)
- EKS gearloan-eks on Kubernetes 1.37, public API endpoint
- Add-ons: vpc-cni, coredns, kube-proxy, pod identity, EBS CSI driver
- terraform.tfvars.example only; state and credentials never in Git

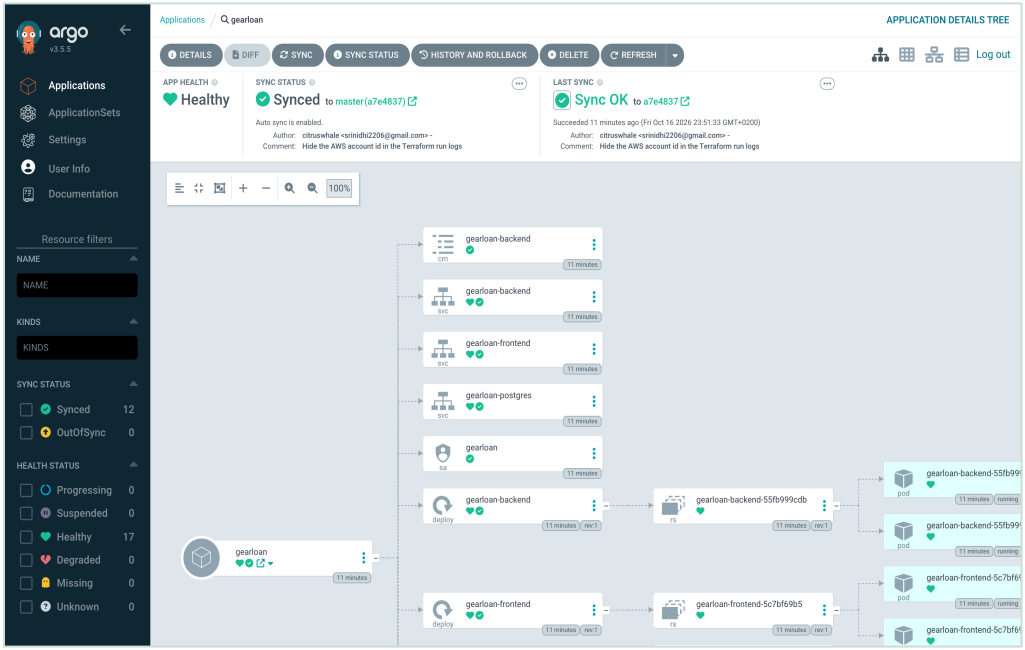
Kubernetes and Helm

Deployments	backend + frontend, 2+ replicas each
Services	ClusterIP for both; headless for Postgres
Ingress	/ → frontend · /api → backend
StatefulSet	PostgreSQL with a PVC on gp3
HPA	CPU 70%, 2–5 Pods; owns the replica count
Probes	startup · readiness /ready · liveness /health
Config	ConfigMap + Secret (password not in Git)



app over the internet through the Ingress + kubectl, on EKS

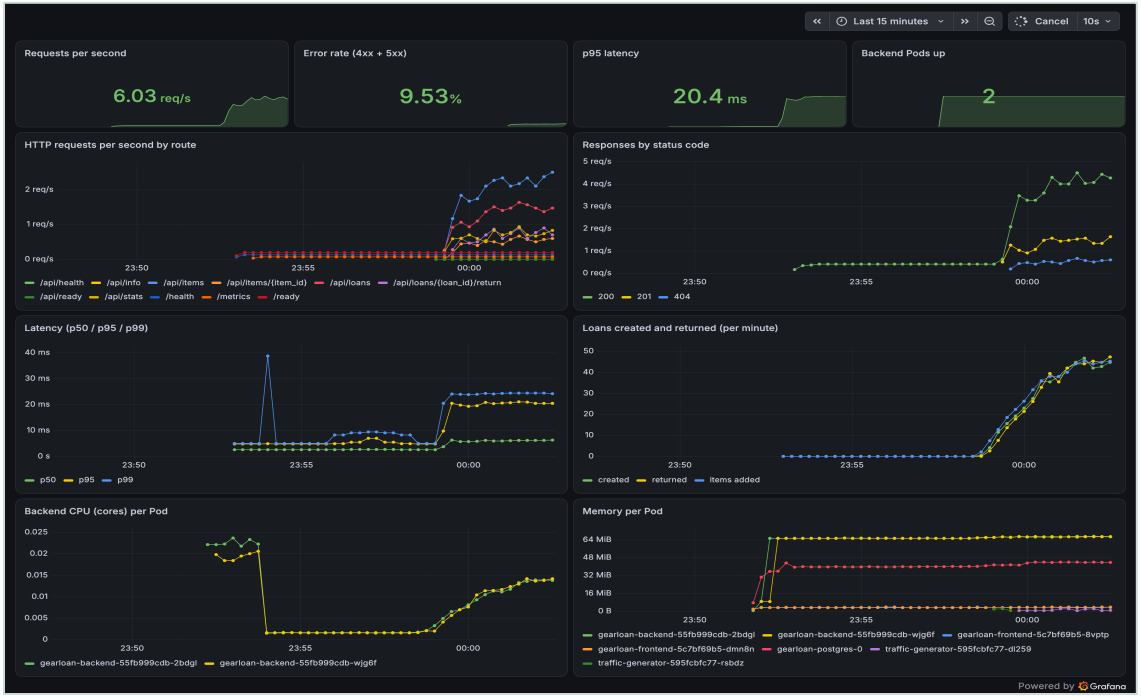
GitOps with Argo CD



Argo CD on EKS: gearloan Synced and Healthy

- Git is the source of truth: the pipeline only writes the new image tag
- Argo CD pulls the Helm chart and keeps the cluster in sync
- prune + self-heal: manual kubectl changes are undone
- CI never needs cluster credentials

Observability: Prometheus and Grafana



my GearLoan dashboard, live on EKS

/metrics

requests by route and status, latency histogram, loans created and returned

ServiceMonitor

Prometheus scrapes both backend Pods: 2/2 UP

Alerts

backend down · >20% 4xx · p95 > 500 ms. The 4xx alert fired and resolved in testing

Tests and local run

28

pytest tests, all passing

4

test files: health, items, loans, migrations

3

containers in Compose

backend — pytest -v

rendered from captured terminal output · Srinidhi Narendran (24BCS10037)

```
$ pytest -v     # 2026-10-16 22:37 CEST, test DB: gearloan_test on localhost:5433 (docker-compose.test.yml)
========================== test session starts ==========================
platform darwin -- Python 3.12.15, pytest-9.1.1, pluggy-1.7.0 -- ~/DevOps SST/Shell_Scripting_Basics/session21-python/backend/.venv/bin/python
cachedir: .pytest_cache
rootdir: ~/DevOps SST/Shell_Scripting_Basics/session21-python/backend
configfile: pytest.ini
testpaths: tests
plugins: anyio-4.15.1
collecting ... collected 28 items

tests/test_health.py::test_health_is_ok PASSED                          [  3%]
tests/test_health.py::test_ready_checks_database PASSED                 [  7%]
tests/test_health.py::test_ready_returns_503_when_db_is_down PASSED     [ 10%]
tests/test_health.py::test_info_shows_name_and_version PASSED           [ 14%]
tests/test_health.py::test_metrics_use_route_templates PASSED           [ 17%]
tests/test_health.py::test_loan_counters_go_up PASSED                   [ 21%]
tests/test_items.py::test_create_and_get_item PASSED                    [ 25%]
tests/test_items.py::test_create_item_validation_errors PASSED          [ 28%]
tests/test_items.py::test_serial_number_must_be_unique PASSED           [ 32%]
tests/test_items.py::test_update_item PASSED                            [ 35%]
tests/test_items.py::test_missing_item_returns_404 PASSED               [ 39%]
tests/test_items.py::test_delete_item PASSED                            [ 42%]
tests/test_items.py::test_search_and_filters PASSED                     [ 46%]
tests/test_items.py::test_items_persist_across_sessions PASSED          [ 50%]
tests/test_loans.py::test_lend_item_marks_it_on_loan PASSED             [ 53%]
tests/test_loans.py::test_loan_validation PASSED                        [ 57%]
tests/test_loans.py::test_loan_for_missing_item_is_404 PASSED           [ 60%]
tests/test_loans.py::test_cannot_lend_item_twice PASSED                 [ 64%]
tests/test_loans.py::test_cannot_lend_item_in_maintenance PASSED        [ 67%]
tests/test_loans.py::test_borrower_limited_to_three_active_loans PASSED  [ 71%]
tests/test_loans.py::test_return_flow PASSED                            [ 75%]
tests/test_loans.py::test_damaged_return_sends_item_to_maintenance PASSED [ 78%]
tests/test_loans.py::test_cannot_delete_item_with_active_loan PASSED    [ 82%]
tests/test_loans.py::test_delete_loan_only_when_returned PASSED         [ 85%]
tests/test_loans.py::test_overdue_status_and_filters PASSED             [ 89%]
tests/test_loans.py::test_stats PASSED                                  [ 92%]
tests/test_migrations.py::test_migration_created_expected_tables PASSED  [ 96%]
tests/test_migrations.py::test_models_match_migration PASSED            [100%]

========================== 28 passed in 0.50s ===========================
```

Tests use a separate gearloan_test database and refuse to run on any other

What I learned

1. **Wait, don't guess** — Services, probes and HPA metrics need a moment; use readiness and rollout status
2. **Size for real use** — Grafana 13 was OOMKilled until it had ~1 GiB
3. **Build for the target** — Mac images are arm64; EKS nodes are x86, so CI builds amd64
4. **One owner per field** — GitOps and the HPA must not both set replicas
5. **Clean up the cloud** — every lab ends with terraform destroy and a leftover check